

An Exact Odd-Level Metaplectic Family for a Toral Hénon Map and a Growing No-Action-Alias Window

Hénon Route-A Working Series, C131

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Abstract

For the toral Hénon automorphism generated by $A = \begin{pmatrix} 3 & -1 \\ 1 & 0 \end{pmatrix}$, we extend a single level-seven quantization to every odd integer $N \geq 3$. One frozen half-phase, Fourier, and chirp convention gives an explicit unitary on $\mathbb{C}^N$, exact Egorov covariance for all N^2 Weyl observables, the unchanged discrete clock, and a uniform antiunitary reversal. An exact integer recurrence also gives a growing interval in which reduction modulo N cannot alias the classical action to the identity. Nine prime and composite certificate levels check 25,313 Weyl cases. The theorem is per-level: it asserts neither cross-level projective compatibility nor a semiclassical trace match.

1 Progress beyond one finite level

An exact lift at one modulus can be a phase accident. The advance here is a single theorem quantified over every odd modulus, including composite ones, with the same signs and reversor. A second theorem makes the pre-alias range explicit and arbitrarily long as the level grows. Neither statement compares finite quantum traces with an external spectrum.

2 Frozen family

Let $N \geq 3$ be odd, write $\omega_N = e^{2\pi i/N}$, and let $h_N \in \mathbb{Z}/N\mathbb{Z}$ satisfy $2h_N = 1$. On the standard basis of $\mathcal{H}_N = \mathbb{C}^N$, set

$$\begin{aligned} Q_N|x\rangle &= \omega_N^x|x\rangle, & P_N|x\rangle &= |x+1\rangle, \\ W_N(q,p) &= \omega_N^{-h_N qp} Q_N^q P_N^p, & (\mathcal{F}_N)_{xy} &= N^{-1/2} \omega_N^{xy}, \\ (C_N)_{xx} &= \omega_N^{3h_N x^2}, & U_N &= C_N \mathcal{F}_N^{-1}. \end{aligned}$$

All residues in operator exponents are modulo N . The orthogonality argument below uses only that ω_N is a primitive N th root; primality of N is never assumed.

Theorem 1 (all-odd exact lift). *For every odd $N \geq 3$, U_N is unitary and*

$$U_N W_N(q,p) U_N^{-1} = W_N(3q-p, q)$$

for all $(q,p) \in (\mathbb{Z}/N\mathbb{Z})^2$. Hence U_N^n implements A^n after exactly n steps. If K is coefficientwise conjugation and $\Theta_N = \mathcal{F}_N K$, then

$$\Theta_N^2 = I, \quad \Theta_N U_N \Theta_N^{-1} = U_N^{-1}, \quad \Theta_N W_N(q,p) \Theta_N^{-1} = W_N(p,q).$$

Proof. Finite Fourier orthogonality proves that $\mathcal{F}_N$ is unitary, and C_N is diagonal unitary. The Weyl product is

$$W_N(v)W_N(w) = \omega_N^{h_N(v_1w_2 - v_2w_1)} W_N(v + w).$$

Direct conjugation gives $U_N Q_N U_N^{-1} = W_N(3, 1)$ and $U_N P_N U_N^{-1} = W_N(-1, 0)$. In a general $W_N(q, p)$, its prefactor $\omega_N^{-h_N qp}$ cancels the product phase $\omega_N^{h_N qp}$ from $W_N(3q, q)W_N(-p, 0)$, proving Egorov without a scalar repair.

Since $\overline{\mathcal{F}_N} = \mathcal{F}_N^{-1}$ and $\overline{C_N} = C_N^{-1}$, $(\mathcal{F}_N K)^2 = I$ and

$$\mathcal{F}_N \overline{U_N} \mathcal{F}_N^{-1} = \mathcal{F}_N C_N^{-1} = U_N^{-1}.$$

Finally $KW_N(q, p)K = W_N(-q, p)$; Fourier conjugation and one Weyl reordering give $W_N(p, q)$. $\square$

3 A growing no-action-alias window

Define $u_0 = 0$, $u_1 = 1$, and $u_{n+1} = 3u_n - u_{n-1}$. For an integer matrix M , write $\|M\|_{\max} = \max_{i,j} |M_{ij}|$.

Proposition 1 (integer pre-alias certificate). *For every $n \geq 1$,*

$$A^n = \begin{pmatrix} u_{n+1} & -u_n \\ u_n & -u_{n-1} \end{pmatrix}, \quad \|A^n - I\|_{\max} = u_{n+1} - 1.$$

If $N > u_{L+1} - 1$, then $A^n \not\equiv I \pmod{N}$ and U_N^n is not scalar for every $1 \leq n \leq L$.

Proof. The matrix formula follows by induction from the recurrence. Positivity and monotonicity show that the upper-left entry of $A^n - I$ has maximal absolute value. If its max norm is below N , congruence to zero modulo N would make every entry literally zero, contradicting $u_n > 0$. A scalar U_N^n has trivial adjoint action; exact Egorov and injectivity of the Weyl labels would then force $A^n \equiv I \pmod{N}$. $\square$

Selected exact receipts make the growth visible:

N	3	9	23	57	145
certified window L	1	2	3	4	5

This is an adjoint-action window, not a trace asymptotic.

4 Exact certificate and controls

The receipt covers levels 3, 5, 7, 9, 11, 15, 23, 57, 145, including composites, and binds 25,313 Egorov cases plus the corresponding antiunitary Weyl cases by case-ledger hashes. An independent implementation reconstructs every digest, threshold, residue, convention, and nonclaim. SymPy checks 50,670 exact identities; twenty-six repaired-hash mutations attack phase signs, reversors, windows, labels, and scope.

At even N , $2h = 1 \pmod{N}$ has no solution, so this literal $qp/2$ convention cannot be copied unchanged. This is not a no-go theorem for doubled-phase or other separately defined even-level Weil/metaplectic conventions. Likewise, the per-level theorem supplies no preferred relative phase between levels.

5 Route-A boundary

The named Hilbert spaces, exact unitaries, observable covariance, clock, and antiunitary justify

(A1_WEAK, A2_FAIL, A3_FAIL, A4_NATURAL_QUANTIZATION).

There is no target divisor census, analytic target completion, cross-level projective theorem, or semiclassical trace match. Route B is unauthorized. The active scope is

NO_BAD_EULER_OR_ROOT_NUMBER.

6 Conclusion

C131 replaces an isolated finite lift by an exact all-odd family and supplies a growing interval free of adjoint-action aliasing. The next separate task would need a rigorously normalized cross-level or semiclassical structure; it is not inferred here.